

## Linear Approximation of the Curve

To simplify the stress-strain curve, we approximate it using 4 linear segments:

1. Segment 1: (0, 0) to (0.01, 43 ksi) - Young's Modulus.
2. Segment 2: (0.01, 43 ksi) to (0.06, 44 ksi) - Plastic.
3. Segment 3: (0.06, 44 ksi) to (0.175, 60 ksi) - Strain hardening.
4. Segment 4: (0.175, 60 ksi) to (0.26, 52 ksi) - Necking.

### Variables to be Used in the Program

| Variable               | Description                                  |
|------------------------|----------------------------------------------|
| strain                 | User inputted strain                         |
| stress                 | Outputted stress                             |
| seg1_slope             | Slope for elastic region                     |
| seg1, seg2, seg3, seg4 | <b>Equations for each respective segment</b> |
| limit                  | <b>limit of strain for this graph</b>        |

strain - User input: strain (x-axis value)

stress - Output: stress (y-axis value)

seg1\_slope - Slope for the elastic region (Segment 1)

seg1, seg2, seg3, seg4 - Equations for the linear segments

limit - limit of strain for this graph (0.26 ksi)

### Steps to Be Followed in the Program

1. Prompt the user to input a value for strain.
2. Determine which segment of the curve the input is positioned at:
  - Strain in Segment 1: Linear elastic (0 to 0.01)
  - Strain in Segment 2: Plastic region (0.01 to 0.02)
  - Strain in Segment 3: Strain hardening (0.02 to 0.04)

- Strain in Segment 4: Necking (0.04 to 0.07)

3. Use linear interpolation for the selected segment to calculate stress.
4. Display the stress corresponding to the user inputted strain.

### Conditionals for the Program

If strain  $< 0.01$ , calculate stress using Young's Modulus (Elastic Region).

If  $0.01 \leq \text{strain} < 0.06$ , interpolate stress using the linear equation for Segment 2.

If  $0.06 \leq \text{strain} < 0.175$ , interpolate stress using the equation for Segment 3.

If  $0.175 \leq \text{strain} < 0.26$ , interpolate stress using the equation for Segment 4.

If strain  $\geq 0.26$ , output message: "Strain exceeds fracture point."

### Test Cases

| Test Case | Strain | Stress                         | Region  |
|-----------|--------|--------------------------------|---------|
| Case 1    | 0.05   | 43.80                          | Elastic |
| Case 2    | 0.08   | 46.78                          | Plastic |
| Case 3    | 0.18   | 59.53                          | Strain  |
| Case 4    | 0.25   | 52.94                          | Necking |
| Case 5    | 0      | 0.00                           | Edge    |
| Case 6    | 0.27   | Strain exceeds fracture point. | Edge    |